

Short-interval higher-order uniformity as a bridge between bounded prime gaps and long prime progressions

Executive summary

The Maynard–Tao bounded-gaps program and the Green–Tao long-arithmetic-progressions program attack *different correlation regimes* of the primes. Bounded gaps are driven by **two-point / second-moment** detection: one optimizes nonnegative Selberg/GPY/Maynard sieve weights to force at least two primes among a finite admissible set of shifts, with the main analytic input being **distribution of primes in arithmetic progressions** at (or beyond) Bombieri–Vinogradov strength. ¹

Long arithmetic progressions (and, more broadly, many finite-complexity linear systems in primes) are **k-point / higher-order uniformity** problems: they require control of multilinear averages of the prime indicator. Green–Tao’s method uses the **W-trick**, a **pseudorandom majorant**, **transference** (relative Szemerédi), and the structure theory of higher Gowers norms via **nilsequences** and inverse theorems. ²

The most promising modern “bridge” between these worlds is now a *specific, technical frontier*: **short-interval higher-order uniformity** for arithmetic functions—especially the von Mangoldt function—measured against nilsequences and expressed via **local Gowers norms**. This line, developed by Kaisa Matomäki ³ and Maksym Radziwiłł ⁴ and collaborators, culminates in results that control nilsequence correlations and Gowers uniformity of (normalized) von Mangoldt in intervals $(x, x + H]$ for wide ranges of H , including “almost all intervals” down to $H \geq X^{1/3+\varepsilon}$, and applications that resemble **Hardy–Littlewood k-tuple asymptotics with a short average over one variable**. ⁵

Conceptually, this frontier is compelling because it is simultaneously: - **sieve-native** (its proofs use Heath–Brown identities, Type I/II sums, bilinear structures—core sieve technology), and - **nilsequence-native** (it treats the true higher-order obstructions identified by Gowers inverse theorems and Möbius–nilsequence orthogonality). ⁶

Analytically, it suggests concrete paths for progress that are neither “pure sieve” nor “pure transference,” but instead aim to import **local higher-order pseudorandomness** into the evaluation/optimization of sieve weights and into localized prime-pattern counting. ⁷

Quantitatively, the best fully verified unconditional bound for bounded gaps remains $H_1 \leq 246$ from D. H. J. Polymath ⁸ (Polymath8b). ⁹ A 2025 ResearchGate manuscript claims $H \leq 234$, but it does not appear (from the evidence located here) in a standard archival venue like arXiv or a peer-reviewed journal; it should be treated as **unverified**. ¹⁰

Programs, correlation order, and toolchains

Core goals

Bounded gaps (Maynard–Tao / Polymath8 lineage). Show $\liminf(p_{n+1} - p_n)$ is finite, and shrink explicit bounds. The foundational breakthroughs are due to Yitang Zhang ¹¹ (first explicit finiteness $< 7 \times 10^7$) and James Maynard ¹² (new sieve refinement implying $\liminf(p_{n+m} - p_n) < \infty$ for all m , and $H_1 \leq 600$); Polymath8b later achieved $H_1 \leq 246$ unconditionally. ¹³

Long arithmetic progressions (Green–Tao). Prove primes contain arbitrarily long arithmetic progressions; more broadly, derive asymptotics for many linear systems in primes (finite complexity, non-degeneracy). This requires a transference framework and higher-order uniformity control (nilsequences). ¹⁴

Correlation order and what is actually being controlled

A helpful formal distinction is the *order of correlation* needed to close the main estimates.

- **Bounded gaps:** essentially **two-point / second-moment** detection. Even though one chooses a k -tuple of shifts $\mathcal{H}$, the analytic core is a quadratic optimization of weights and evaluation of sums involving $\Lambda(n)$ (or $\theta(n)$) along shifts—controlled primarily by prime distribution in arithmetic progressions and bilinear/Type I/II estimates. ¹⁵
- **Long APs:** truly **k-point / higher-order**. Counting k -term APs involves k -fold products $f(n)f(n+d) \cdots f(n+(k-1)d)$, governed by Gowers U^{k-1} norms. Large Gowers norms correspond to correlations with structured objects (nilsequences), by inverse theorems. ¹⁶

Comparison table

Attribute	Bounded gaps (Maynard–Tao / Polymath8b)	Long APs / linear patterns (Green–Tao and beyond)
Problem	Infinitely often $p_{n+1} - p_n \leq H$; optimize H . ¹⁷	Existence of k -term APs for all k ; asymptotics for many linear systems. ¹⁸
Correlation order	2-point (second-moment) core, despite many shifts. ¹⁹	k -point; controlled by U^{k-1} norms and nilsequence obstructions. ²⁰
Main tools	Selberg/GPY/Maynard sieve weights; admissible tuples; BV-type distribution; Type I/II sums. ²¹	W-trick; pseudorandom majorant; transference/relative Szemerédi; Gowers norms; nilsequences; inverse theorems. ²²
Typical analytic inputs	Bombieri–Vinogradov and refinements; stronger distribution inputs improve constants. ²³	Möbius–nilsequence orthogonality; inverse theorems; nilsequence equidistribution; majorant linear forms estimates. ²⁴

Attribute	Bounded gaps (Maynard–Tao / Polymath8b)	Long APs / linear patterns (Green–Tao and beyond)
Representative unconditional quantitative bounds	Zhang: $H_1 < 7 \times 10^7$. ²⁵ Maynard: $H_1 \leq 600$. ¹⁹ Polymath8b: $H_1 \leq 246$. ²⁶	Green–Tao: APs of all lengths (qualitative). ²⁷ Short-interval quantitative pattern counts now exist (see below). ²⁸
Conditional benchmarks	Under GEH: Polymath8b gets $H_1 \leq 6$. ²⁶ Under EH: Maynard notes strong improvements (e.g. $H_1 \leq 12$). ²⁹	Many results unconditional after GI(s) and Möbius–nilsequence theorems; quantitative dependence remains huge. ²⁰
Open problems	Twin primes $H_1 = 2$; parity barriers; turn “bounded gap exists” into “specific gap occurs infinitely often.” ¹⁷	Effective quantitative bounds; extend local uniformity to much shorter intervals; treat more degenerate systems. ³⁰

Milestones and the emergence of the short-interval bridge

Mermaid timeline of key milestones

<pre> timeline title Milestones toward the short-interval "bridge" (as of 2026-02-28) 2004 : Green-Tao preprint: transference + GPY/Goldston-Yildirim majorant ideas 2008 : Green-Tao (Annals): arbitrarily long APs in primes 2010 : Green-Tao: linear equations in primes (finite complexity framework) 2012 : Green-Tao-Ziegler (Annals): inverse theorem for Gowers norms (nilsequences) 2012 : Green-Tao (Annals): Möbius strongly orthogonal to nilsequences 2013 : Zhang: first finite bound on H1 (bounded gaps) 2014 : Polymath8b: H1 ≤ 246 unconditional; ≤ 6 under generalized EH 2015 : Maynard (Annals): refined GPY sieve; H1 ≤ 600; Hm finite for all m 2016 : Matomäki-Radziwiłł (Annals): multiplicative functions in almost all short intervals 2020 : Matomäki-Radziwiłł-Tao-Teräväinen-Ziegler (arXiv→Annals 2023): higher uniformity for bounded multiplicative functions in short intervals on average 2022 : Higher uniformity of arithmetic functions in short intervals I (Λ, μ, dk), “all intervals” regime 2024 : Higher uniformity of arithmetic functions in short intervals II (Λ, μ, dk), “almost all intervals” regime (v2 in 2026) 2026 : Published (Inventiones): short-interval Gowers uniformity for Λ/μ and k-tuple asymptotics with one short averaging variable 2025 : Unverified ResearchGate claim: H ≤ 234 (not in archival venue) </pre>

The core message of this trajectory is that “short-interval uniformity” progressed in layers: 1. cancellation for multiplicative functions in almost all short intervals; ³¹

2. higher-order (nilsequence/Gowers) uniformity for bounded multiplicative functions on average; ³²
3. higher-order uniformity for von Mangoldt itself (after subtracting a structured approximant), first in a relatively long “all intervals” regime, then in an “almost all intervals” regime reaching $H \geq X^{1/3+\varepsilon}$, with explicit k-tuple-type applications. ⁵

This is precisely the bridge zone: it treats Λ with tools that look simultaneously like sieve (Type I/II) and like higher-order Fourier analysis (nilsequences).

The short-interval higher-order uniformity frontier

The “average” and “almost all” regimes for higher-order uniformity

The 2023 Annals paper by Matomäki–Radziwiłł–Tao–Teräväinen–Ziegler proves that for large X and H in the range $X^\theta \leq H \leq X$ (for any fixed $\theta > 0$), one has **on average over** $x \in [X, 2X]$ strong cancellations of non-pretentious 1-bounded multiplicative functions against polynomial phases and even **nilsequences**, and correspondingly small local Gowers norms. ³³

This is already philosophically aligned with Green–Tao: “structured obstructions are nilsequences; if you beat them, you have uniformity.” The gap was that this result is for *bounded multiplicative functions* (e.g. Liouville), not Λ .

That gap is narrowed by the “Higher uniformity of arithmetic functions in short intervals” series:

- **Part I (All intervals).** For intervals $(X, X + H]$ with H in a range $X^{\theta+\varepsilon} \leq H \leq X^{1-\varepsilon}$, it studies higher uniformity of μ , Λ , and divisor functions d_k , by subtracting structured approximants $\Lambda^\sharp, d_k^\sharp$ (and $\mu^\sharp = 0$) and showing the difference has small nilsequence correlations and hence small local Gowers norms in certain regimes (notably $\theta = 5/8$ for $f \in \{\Lambda, \mu, d_k\}$ as an example in the abstract). ³⁴
- **Part II (Almost all intervals).** For *almost all* $x \in [X, 2X]$, it controls nilsequence correlations of $\Lambda - \Lambda^\sharp$ in intervals $(x, x + H]$ for significantly shorter H , with the abstract highlighting $H \geq X^{1/3+\varepsilon}$ for Λ (and even $H \geq X^\varepsilon$ for divisor functions with appropriate normalization). ³⁵

The 2026 published Inventiones version makes this “bridge” even more explicit: it states short-interval **Gowers uniformity estimates** (Theorem 1.3) and then derives an ℓ -**point Hardy–Littlewood asymptotic with only one averaging variable short-averaged** (Theorem 1.5), valid for a density $1 - o(1)$ of shifts $h \leq H$ when $H \geq X^{1/3+\varepsilon}$. ³⁶

Why this is a *bridge* to bounded gaps (not merely a Green–Tao refinement)

The bounded-gaps sieve needs to evaluate weighted sums of the form

$$\sum_{n \in I} w(n) \Lambda(n + h_i) \quad \text{and} \quad \sum_{n \in I} w(n),$$

often with $I = [x, 2x]$ in the classical global setting. What makes the short-interval bridge plausible is that the Part I/II program for Λ is **built from the same analytic primitives** that appear when controlling sieve weights: - Heath–Brown identities / decompositions of Λ into convolutions;
 - Type I / Type II bilinear sums;
 - distributional control against structured objects. ³⁷

Part II describes new tools—e.g. a “nilsequence contagion lemma” to scale up control from smaller to larger scales—precisely the kind of device that could, in principle, interface with **well-factorable-weight distribution theorems** central to modern improvements of distribution beyond $x^{1/2}$. ³⁸

Representative theorems and quantitative bounds you can “hang your hat on”

Bounded gaps (verified): - Zhang: $\liminf(p_{n+1} - p_n) < 7 \times 10^7$. ²⁵

- Maynard: refinement of GPY; in particular $H_1 \leq 600$ and $H_m < \infty$ for all m . ¹⁹

- Polymath8b: $H_1 \leq 246$ unconditionally, and $H_1 \leq 6$ under generalized Elliott–Halberstam. ⁹

Unverified bounded-gap claim (flagged): - A 2025 ResearchGate upload claims $H \leq 234$ via a new “weighted distribution” framework, but it is not located in an archival venue here; treat as unverified. ¹⁰

Long APs and the higher-order framework (verified): - Green–Tao: primes contain arbitrarily long arithmetic progressions (Annals 2008). ²⁷

- Inverse theorem for Gowers norms: large U^{s+1} implies correlation with an s -step nilsequence (Annals 2012), with later quantitative refinements. ²⁰

- Möbius strongly orthogonal to nilsequences (Annals 2012). ³⁹

Short-interval higher-order uniformity (the bridge): - “Higher uniformity ... on average” for bounded multiplicative functions in short intervals (Annals 2023). ³²

- “Higher uniformity of arithmetic functions in short intervals I” (Forum of Mathematics, Pi 2023; arXiv v4 2024): nilsequence correlation bounds for $\Lambda - \Lambda^\sharp$ in deterministic $(X, X + H]$ with H as low as $X^{5/8+\varepsilon}$ in the exemplar regime, yielding short-interval linear-equation counts in primes. ³⁴

- “Higher uniformity ... II / Inventiones 2026”: for almost all intervals and $H \geq X^{1/3+\varepsilon}$, establishes Gowers uniformity for (normalized) von Mangoldt after W-trick and derives ℓ -point Hardy–Littlewood with one short averaging variable for most shifts $h \leq H$. ⁴⁰

This last item is the strongest current evidence that local higher-order uniformity is not just a philosophical bridge but a **technical conduit** from sieve decompositions to nilsequence control and then back to prime-pattern asymptotics.

How the bridge could connect sieve optimization and nilsequence control

Mermaid flowchart comparing method pipelines with the bridge highlighted

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flowchart TD
    subgraph MT["Bounded gaps pipeline (Maynard–Tao / Polymath8)"]
    end
  
```

```

    MT1["Choose admissible k-tuple H"] --> MT2["Build Selberg/GPY/Maynard weight
w(n) ≥ 0"]
    MT2 --> MT3["Evaluate  $\sum w(n)$  and  $\sum w(n)\Lambda(n+h_i)$  using primes-in-AP + Type I/
II"]
    MT3 --> MT4["Optimize quadratic functional / variational problem"]
    MT4 --> MT5["Deduce  $\geq 2$  primes among shifts infinitely often  $\rightarrow$  explicit H"]
end

subgraph Bridge["Bridge substrate: short-interval higher-order uniformity"]
    B1["Heath-Brown identity  $\rightarrow$  decompose  $\Lambda$  in short interval"] --> B2["Type I/II
bilinear control + nilsequence estimates"]
    B2 --> B3["Control of  $\Lambda\text{-}\Lambda^\#$  against nilsequences in  $(x, x+H)$ "]
    B3 --> B4["Local Gowers norms small; local pseudorandom majorants via W-
trick"]
end

subgraph GT["Long AP pipeline (Green-Tao and extensions)"]
    GT1["W-trick: restrict to  $Wn+b$ "] --> GT2["Majorant  $v$  for primes; linear
forms conditions"]
    GT2 --> GT3["Decompose into structured (nilsequence) + uniform part
(Gowers)"]
    GT3 --> GT4["Transference / relative Szemerédi  $\rightarrow$  k-term APs and other
patterns"]
end

MT3 --> Bridge
Bridge --> GT2

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This picture encodes a precise hypothesis: the bridge becomes genuinely unifying if one can take the short-interval control (B3–B4) and use it to improve MT3 in regimes that matter for sieve optimization, while simultaneously allowing GT2–GT4 to be localized into short intervals with effective uniformity conditions.

Why short-interval uniformity is particularly promising

Short intervals force both programs to confront the same bottlenecks: - In bounded gaps, localizing to $[x, x + H]$ demands prime distribution and bilinear estimates with fewer averaging degrees of freedom; Alweiss–Luo’s $\delta \geq 0.525$ threshold comes from the best known *uniform* prime number theorem technology in such lengths. ⁴¹

- In Green–Tao-type counting, localizing “pseudorandomness” requires local Gowers norms and nilsequence correlations of arithmetic functions; the Matomäki–Radziwiłł program provides exactly these estimates (first for multiplicative functions, now for Λ). ⁴²

The Inventiones paper explicitly notes that quantifying these Gowers bounds is delicate because it would require tracking the “nilsequence contagion” argument’s dependence on dimension and complexity, indicating a clear technical obstacle and a clear direction for sharpening. ³⁶

Research directions prioritized on the bridge

The directions below are phrased as “analytic paths” rather than broad speculation. Each item states (1) feasibility, (2) partial results, (3) main obstacles, and (4) what new inputs would likely be required.

Local Maynard sieve powered by short-interval uniformity of von Mangoldt

Idea. Replace the global interval $[x, 2x]$ in Maynard’s method by short intervals $(x, x + H]$, and attempt to prove *dense* bounded-gap phenomena in most such intervals using the Matomäki–Radziwiłł–Shao–Tao–Teräväinen uniformity of $\Lambda - \Lambda^\sharp$.

Feasibility. Moderately plausible in an “almost all intervals” sense. The bounded-gaps method already has a short-interval adaptation at exponent 0.525 (Alweiss–Luo), but that relies on (uniform) prime-density technology rather than higher-order uniformity. ⁴¹ The new uniformity results reach $H \geq X^{1/3+\varepsilon}$ for almost all intervals in the nilsequence/Gowers sense, which is substantially shorter. ⁴³

Known partial results. - Alweiss–Luo: for any $\delta \in [0.525, 1]$, there exist k, d such that $[x - x^\delta, x]$ contains $\gg x^\delta / (\log x)^k$ pairs of consecutive primes with gap $\leq d$ for all sufficiently large x . ⁴¹
- Inventiones 2026: for $H \geq X^{1/3+\varepsilon}$, obtains k -tuple-type asymptotics with only one short averaging variable for most shifts $h \leq H$, which is “prime tuples on average over shifts” in a genuinely short range. ³⁶

Main obstacles. - Bounded gaps needs **positivity and existence** for explicit finite sets of shifts; the Inventiones-type results often give asymptotics for “most” shifts or “almost all” intervals. Bridging from averaged asymptotics to guaranteed existence in each interval is nontrivial and may reintroduce parity-type barriers. ⁴⁴

- The sieve weight $w(n)$ is itself structured (divisor sums), and one must control correlations of Λ with those weights in short intervals. The uniformity statements in Part II control $\Lambda - \Lambda^\sharp$ against nilsequences, which is powerful but not immediately tailored to arbitrary sieve weights. ⁴³

Required new inputs. - **Effective nilsequence decompositions for sieve weights** (or a proof that the relevant class of Maynard weights can be reduced to combinations of nilsequence tests plus acceptable error).

- Stronger **short-interval distribution in arithmetic progressions** for the moduli induced by the weight divisors, ideally with well-factorable structure.

- Better quantitative control of inverse theorems (complexity dependence), to translate “uniform against nilsequences” into explicit error terms in weighted sums. ⁴⁵

Plausibility + difficulty. High conceptual plausibility; very hard technically. The “right” intermediate target is likely: *bounded gaps in almost all intervals of length X^θ for some $\theta < 0.525$* , rather than improving the global H_1 bound immediately.

Short-interval level of distribution for well-factorable weights as the interface layer

Idea. Use the “factorability” paradigm—central in distribution beyond the square-root barrier—to connect Maynard-style sieve evaluation with nilsequence-based control mechanisms in short intervals.

Why this is on the bridge. “Well-factorable” and “triply well-factorable” weights are exactly where modern distribution of primes in AP has exceeded $x^{1/2}$, and these results are explicitly designed to feed sieve applications. ⁴⁶

Known partial results. - Maynard (2020): primes in AP to moduli up to $x^{3/5-\varepsilon}$ when summed with suitably well-factorable weights. ⁴⁷

- Lichtman (2023): level of distribution $66/107 \approx 0.617$ using triply well-factorable weights, with conditional extension to $5/8$ under Selberg eigenvalue conjecture. ⁴⁸

- Pascadi (2025): unconditional equidistribution up to $x^{5/8-o(1)}$ via triply well-factorable weights (and improvements for linear sieve weights), removing Selberg-eigenvalue dependence in earlier work. ⁴⁹

These are not “short interval” results per se, but they represent the strongest current analytic control of primes against structured moduli weights, and therefore define the most realistic input channel to sieve optimization.

Main obstacles. - Translating global mean-value theorems in AP to **short-interval** statements is delicate: short-interval AP distribution typically loses averaging power and requires distinct methods (zero-density bounds, short-interval dispersion, etc.).

- The short-interval uniformity papers for Λ rely heavily on Type I/II estimates and nilsequence arguments; incorporating factorable weight distribution into that framework requires new hybrid bilinear forms technology (essentially “factorable Type II meets nilsequence contagion”). ⁵⁰

Concrete analytic path. 1. Prove a *short-interval* analogue of “primes in AP to large moduli with factorable weights,” at least averaged over $x \in [X, 2X]$.

2. Use that to evaluate Maynard weights localized to $(x, x + H]$ with smaller H , improving Alweiss–Luo-type thresholds or increasing the density of bounded gaps in short windows. ⁵¹

Required new inputs. - Short-interval versions of the spectral large sieve bounds that underpin the best “beyond $1/2$ ” distribution results, with uniformity in the shift/interval parameter. ⁵²

- Mechanisms to fuse factorability with nilsequence-test amplification (potentially inspired by the Inventiones “contagion lemma” philosophy). ³⁶

Plausibility + difficulty. Plausible but very high difficulty. This is probably the most realistic route by which the bridge could eventually feed back into improved bounded-gap constants, because it targets the exact bottleneck (distribution input) that controls sieve power.

Local transference in short intervals using von Mangoldt uniformity

Idea. Use short-interval Gowers uniformity of $\Lambda - \Lambda^\sharp$ (after W-trick normalization) to formulate and prove **short-interval transference** statements: within most intervals $(x, x + H]$, primes behave like a dense subset of a pseudorandom measure for purposes of counting finite-complexity configurations.

Known partial results. - Part I already advertises asymptotic formulas for solutions to linear equations in primes in short intervals (within its H range), which is a direct “local Green–Tao-type” statement. ³⁴

- The Inventiones paper’s Theorem 1.5 (“ ℓ -point Hardy–Littlewood with one averaging variable”) is

essentially a local correlation estimate strong enough to underwrite generalized von Neumann arguments in short ranges. ³⁶

Main obstacles. - The local uniformity is proved relative to an approximant $\Lambda^\sharp$ and after a W-trick normalization. Making the majorant pseudorandomness conditions *effective and configuration-stable* (uniform over the full family of linear forms needed for a given combinatorial theorem) is technically demanding. ²⁸

- Quantitative inverse theorems: while qualitative inverse theorems are established, quantitative dependencies remain enormous; some recent work improves these bounds but they are still very large, and the Inventiones paper itself flags dimension-dependence issues for quantification. ⁴⁵

Concrete analytic path. - Target a “finite complexity, one-short-variable averaged” version of Green–Tao’s linear-forms framework in $(x, x + H]$, pushing H lower by improving Type II estimates and the nilsequence contagion machinery.

- Extract explicit configuration counts: e.g., bounded-length APs entirely inside $(x, x + H]$ for most x , with $H = X^\theta$ and explicit θ thresholds. ⁵³

Required new inputs. - Sharper Type II estimates for Λ in short intervals (the source of the $1/3$ threshold in the “almost all” setting). ⁴³

- Better quantitative equidistribution for polynomial nilorbits with explicit complexity dependence (to stabilize error control across families of configurations). ²⁰

Plausibility + difficulty. High plausibility as a continuing program (it is already partially realized). Difficulty is concentrated in lowering the short-interval exponent and improving uniformity from “almost all” to “all” intervals.

Computational experiments and prototypes for testing the bridge

The analytic theorems are asymptotic and often effective only at very large scales. Nonetheless, carefully designed computations can test *model fidelity* (sieve-majorant vs primes), *local correlation behavior*, and the practical meaning of “uniformity” proxies at accessible X .

The experiments below are designed to be implementable with segmented sieves and FFT-based correlation calculations on X up to 10^8 – 10^{10} depending on hardware. (Resource estimates assume an efficient segmented sieve and C/Julia/Rust-level performance.)

Recommended experiments table

Experiment	Purpose (bridge question)	Parameters to sweep	Data sources	Metrics	Expected outcome if “bridge is real”
Local U^2 / Fourier uniformity of primes after W-trick	Does short-interval pseudorandomness look “sieve-like” locally?	$X \in [10^7, 10^9]$; $H \in \{X^{0.25}, X^{0.33}, X^{0.5}\}$; $W \in \{30, 210\}$	Segmented prime sieve	Max nontrivial Fourier coefficient of $1_{\mathbb{P}}(Wn + b)$ – mean per window	Peaks dominated by finite-sieve coupling; decay as H grows; stabilization across residues
Local nilsequence proxy tests (polynomial phases)	Do prime-weighted sums cancel against low-degree phases in short windows?	Same X, H, W ; phases $e(\alpha n)$, $e(\alpha n^2)$ with α rational grid	Primes; optionally Liouville/ Möbius	$\max_{\alpha} \sum_{x < n \leq x+H} (f(n) - f^{\#}(n))e(\alpha n^d) $	Cancellation improves with H ; behavior closer to MRRTZ predictions for bounded mult. functions
Sieve-majorant fidelity in short intervals	Does a finite-sieve majorant predict short-interval pattern counts?	B ladder for “sieve depth”; H as above	Sieve survivors vs actual primes	Ratio of counts: (i) primes in $(x, x + H]$, (ii) prime pairs at gap $\leq d$, (iii) 3AP counts	Ratios approach 1 as sieve depth increases; residual deviations small and structured
Maynard weight $\times$ local patterns (small k prototypes)	Are Maynard weights locally “uniform” against higher-order tests?	Choose small admissible tuples; vary truncation R ; localize to windows	Compute Maynard/ GPY weight $w(n)$ and primes	Compare $\sum w(n)\Delta(n + h_i)$ to model; test correlation of $w(n)$ with polynomial phases	$w(n)$ largely explained by multiplicative structure; residual correlation small after normalization

Experiment	Purpose (bridge question)	Parameters to sweep	Data sources	Metrics	Expected outcome if “bridge is real”
Short-interval bounded gaps frequency vs uniformity proxies	Does local uniformity predict where small gaps occur more/less than expected?	Many windows of fixed H	Primes	Small gap counts; correlate with (i) local Fourier peaks, (ii) local variance of Λ -approx residual	Weak-to-moderate correlation; if strong correlation exists, suggests exploitable structure
“One-short-variable averaged tuples” numerics	Mirror Theorem 1.5 empirically at finite scale	Fix ℓ ; sample $h \leq H$	Primes	Estimate $\sum_{n \leq X} \prod_{j=0}^{\ell-1} \Lambda(n + jh)$ vs singular series heuristic	Agreement improves with X and when averaging over h ; deviations mostly from small primes

Prototype design notes (what to do, not code)

- Use a segmented sieve to generate primes in blocks and compute windowed statistics. For Λ -weighted sums, approximate $\Lambda(n)$ by $\log p$ on primes p and 0 otherwise; for more faithful modeling, include prime powers if desired.
- Implement W-trick by restricting to residue classes $b \bmod W$ with $(b, W) = 1$, and analyze the induced sequence in the n -variable.
- For nilsequence proxies at low step, polynomial phases $e(\alpha n^d)$ are a computationally accessible stand-in; MRRTZ explicitly links such phase tests to higher uniformity results and notes that nilsequences are the correct general obstruction class. ⁵⁴

These experiments will not “verify theorems,” but can (i) validate modeling assumptions (finite-sieve majorant accuracy), (ii) expose which local deviations are fully sieve-explained, and (iii) highlight whether any residual structure correlates with known obstruction classes (polynomial/nilsequence types), which is exactly what the bridge claims should happen.

Notes on bounded-gap status and the unverified $H \leq 234$ claim

The most authoritative, archival statement remains Polymath8b’s **unconditional** $H_1 \leq 246$. ⁹

A 2025 ResearchGate manuscript by “Yuhang Shi” claims $H \leq 234$ via a new weighted distribution framework. Because it is not located here in arXiv/Annals/other standard archival venues, and because the

bounded-gap record is widely tracked in those venues, it should be treated as **unverified until independently vetted.** ¹⁰

In contrast, the short-interval higher-order uniformity results that define the bridge are available in major venues (Annals; Forum of Mathematics, Pi; Inventiones) and on arXiv, and are therefore the most reliable “modern frontier” anchor for research planning. ⁵⁵

¹ ¹⁵ ¹⁹ ²¹ Small gaps between primes - Annals of Mathematics

https://annals.math.princeton.edu/2015/181-1/p07?utm_source=chatgpt.com

² ⁴ ⁸ ¹⁴ ¹⁶ ¹⁸ ²² ²⁷ The primes contain arbitrarily long arithmetic progressions

https://annals.math.princeton.edu/wp-content/uploads/annals-v167-n2-p03.pdf?utm_source=chatgpt.com

³ ⁷ ⁴¹ ⁵¹ Bounded gaps between primes in short intervals

https://arxiv.org/abs/1707.05437?utm_source=chatgpt.com

⁵ ²⁸ ³⁴ [2204.03754] Higher uniformity of arithmetic functions in short intervals I. All intervals

<https://arxiv.org/abs/2204.03754>

⁶ ¹² ³⁰ ³⁶ ³⁷ ⁴⁰ ⁵⁰ ⁵³ Higher uniformity of arithmetic functions in short intervals II. Almost all intervals | Inventiones mathematicae | Springer Nature Link

<https://link.springer.com/article/10.1007/s00222-026-01408-6>

⁹ ¹⁷ ²⁶ ⁴⁴ [1407.4897] Variants of the Selberg sieve, and bounded ...

https://arxiv.org/abs/1407.4897?utm_source=chatgpt.com

¹⁰ (PDF) A Weighted Distribution of Primes and a New ...

[https://www.researchgate.net/publication/393888742_A_Weighted_Distribution_of_Primes_and_a_New_Unconditional_Bound_on_Gaps_Between_Primes?](https://www.researchgate.net/publication/393888742_A_Weighted_Distribution_of_Primes_and_a_New_Unconditional_Bound_on_Gaps_Between_Primes?utm_source=chatgpt.com)

[utm_source=chatgpt.com](https://www.researchgate.net/publication/393888742_A_Weighted_Distribution_of_Primes_and_a_New_Unconditional_Bound_on_Gaps_Between_Primes?utm_source=chatgpt.com)

¹¹ ³⁵ ³⁸ ⁴³ [2411.05770] Higher uniformity of arithmetic functions in short intervals II. Almost all intervals

<https://arxiv.org/abs/2411.05770>

¹³ ²³ ²⁵ Bounded gaps between primes - Annals of Mathematics

https://annals.math.princeton.edu/2014/179-3/p07?utm_source=chatgpt.com

²⁰ An inverse theorem for the Gowers $U^{s+1}[N]$ -norm

https://annals.math.princeton.edu/2012/176-2/p11?utm_source=chatgpt.com

²⁴ ³⁹ The Möbius function is strongly orthogonal to nilsequences

https://annals.math.princeton.edu/2012/175-2/p03?utm_source=chatgpt.com

²⁹ Small gaps between primes - Annals of Mathematics

https://annals.math.princeton.edu/wp-content/uploads/annals-v181-n1-p07-p.pdf?utm_source=chatgpt.com

³¹ ⁴² Multiplicative functions in short intervals - Annals of Mathematics

https://annals.math.princeton.edu/2016/183-3/p06?utm_source=chatgpt.com

³² ⁵⁵ Higher uniformity of bounded multiplicative functions in short ...

https://annals.math.princeton.edu/2023/197-2/p03?utm_source=chatgpt.com

33 54 Higher uniformity of bounded multiplicative functions in short intervals on average

https://arxiv.org/abs/2007.15644?utm_source=chatgpt.com

45 Quantitative bounds in the inverse theorem for the Gowers U^{s+1} -norms over cyclic groups

https://arxiv.org/abs/1811.00718?utm_source=chatgpt.com

46 47 Primes in arithmetic progressions to large moduli II

https://arxiv.org/abs/2006.07088?utm_source=chatgpt.com

48 52 Primes in arithmetic progressions to large moduli, and Goldbach beyond the square-root barrier

https://arxiv.org/abs/2309.08522?utm_source=chatgpt.com

49 On the exponents of distribution of primes and smooth numbers

https://arxiv.org/abs/2505.00653?utm_source=chatgpt.com